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# AI PRACTICAL
# Travelling Salesperson Problem using Heuristic Search
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# STEP 1 : IMPORT LIBRARY

import itertools

# STEP 2 : CREATE DISTANCE MATRIX

cities = {

    'A': {'A':0, 'B':10, 'C':15, 'D':20},

    'B': {'A':10, 'B':0, 'C':35, 'D':25},

    'C': {'A':15, 'B':35, 'C':0, 'D':30},

    'D': {'A':20, 'B':25, 'C':30, 'D':0}

}

# STEP 3 : STORE CITY NAMES

city_names = list(cities.keys())

# STEP 4 : FIND ALL POSSIBLE PATHS

all_paths = itertools.permutations(city_names)

# STEP 5 : FIND MINIMUM COST PATH

min_cost = 9999

best_path = None

for path in all_paths:

    cost = 0

    for i in range(len(path)-1):

        cost += cities[path[i]][path[i+1]]

    cost += cities[path[-1]][path[0]]

    if cost < min_cost:

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min_cost = cost
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best_path = path
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# STEP 6 : DISPLAY OUTPUT
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print("Best Path : ", best_path)
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print("Minimum Cost : ", min_cost)
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